AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM271 Machine Learning and Deep Learning

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 15 /April / 2026

Number of Exam Pages: 2 (including this cover sheet)	**Time Allowed: 120 Minutes**

Instructions:

- Total Marks: 30
- This exam consists of 3 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name** and **ID** on the External Answer booklet.
- **Calculators are not allowed**

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Part – A (5 Marks)

1. In supervised learning, the training data contains
   a) Only input features
   b) No labels
   c) Input data with desired outputs (labels)
   d) Only clusters
2. Which measure is commonly used to evaluate regression models in the lecture slides?
   a) Precision
   b) Recall
   c) RMSE
   d) F1-score
3. Which gradient descent variant computes the gradient using a single training instance at a time?
   a) Batch Gradient Descent
   b) Mini-batch Gradient Descent
   c) Stochastic Gradient Descent
   d) Normal Equation
4. Which of the following is an ensemble learning method?
   a) Random Forest
   b) PCA
   c) K-Means
   d) LLE
5. Which dimensionality reduction method projects data onto directions of maximum variance?
   a) DBSCAN
   b) PCA
   c) Decision Tree
   d) AdaBoost

Part – B (15 Marks)

Provide brief answers to the following questions.

1) Explain the purpose of regularization in machine learning. Differentiate between Ridge Regression and Lasso Regression. (3 Marks)
2) Compare Batch Gradient Descent and Stochastic Gradient Descent. (3 Marks)
3) Explain the purpose of a confusion matrix. Also define precision and recall. (6 Marks)
4) Explain any three commonly used clustering techniques. (3 Marks)

Part – C (10 Marks)

NOTE: This question does not need any programming code.

A real estate company wants to build a machine learning model to predict housing prices using features such as location, number of rooms, population, and median income. As a data scientist, explain how you would build this prediction system using the machine learning pipeline discussed in class.

Your task is to design a complete machine learning workflow to build this system.

Instructions:

- Provide a solution for building a recommendation system.
- Present your solution in a stepwise manner.
- Ensure that your response consists of a minimum of four steps.

End of questions

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